

PC sur mesure

SEG 2505 - Introduction to Software Engineering

Fall 2024

School of Electrical Engineering and Computer Science, University of Ottawa

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GitHub Project Reference :

GitHub Link: <https://github.com/uOttawa-2024-2025-seg2505-projet/groupe-8>

The repository contains:

- **Android Studio** source codes.
- **APK files** for each deliverable.
- **UML diagrams** (activity, class, sequence, state).
- **README files** and deliverable tags.

Design Choices:

Description of major design choices:

Software Architecture:

- **Modular Architecture:** To clearly separate roles and responsibilities (Requester, StoreKeeper, Assembler), a modular architecture was adopted. This allows for better maintainability and scalability of the application by isolating different functionalities into distinct modules.

Database Management:

- **Firebase:** Instead of SQLite, Firebase was chosen as the database solution. Firebase offers several advantages such as:
 - **Real-time Database:** Allows instantaneous data synchronization between clients and the server, facilitating real-time order and stock management.
 - **Secure Authentication:** Firebase Authentication provides robust methods for managing user registration and authentication, strengthening system security.
 - **Scalability:** Firebase is designed to easily adapt to an increase in users and data volume without requiring major architectural changes.
 - **Easy Android Integration:** Firebase integrates seamlessly with Android Studio, simplifying development and data management.

User Interface (UI):

- **Compliance with Android Guidelines:** The user interface was designed following Android design guidelines to ensure a consistent, intuitive, and pleasant user experience. The application is available in French and English to reach a wider audience.

Role Management:

- **Role-Based Access Control:** The application implements access restrictions based on roles (Requester, StoreKeeper, Assembler, Administrator) to secure and organize the functionalities accessible to each type of user. Firebase Authentication facilitates the management of roles and permissions.

Scenario Compliance:

- **Complete Order Cycle:** The main scenario of the application follows the order cycle, from creation to delivery, including stock management and custom PC assembly. Firebase allows for efficient tracking and management of each stage of this cycle thanks to its real-time data management capabilities.

Justifications for Using Firebase:

- **Ease of Development:** Firebase simplifies many backend tasks, such as authentication, database management, and hosting. This allows the team to focus more on developing frontend features.
- **Real-time and Responsiveness:** Firebase's ability to synchronize data in real-time is crucial for an application where order and stock information must be constantly updated and instantly accessible to users.
- **Security:** Firebase offers granular security rules that allow data access control based on user roles. This ensures that only authorized individuals can perform specific actions or access particular information.
- **Scalability and Maintenance:** Firebase is a managed solution that scales with the application's needs. This reduces the effort required for the maintenance and evolution of the backend infrastructure.

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Last Name

First Name

Last Name

Email Address

Email Address

✓

Username

Username

Password

Password

o

Confirm Password

Confirm Password

SIGN UP


uOttawa

Uottawa Hardware

Please login with your information

Email

✓

Password

o

☐ Remember me

Forgot password

LOGIN

or

SIGN UP

Admin: Homepage


PC SUR MESURE
VOTRE PC. VOTRE STYLE

Admin Dashboard

MANAGE STOCK

ADD USERS

REMOVE USERS

CREATE ORDER

MY ORDERS

LOGOUT

Search Users

Item 1	Sub Item 1
Item 2	Sub Item 2
Item 3	Sub Item 3
Item 4	Sub Item 4
Item 5	Sub Item 5
Item 6	Sub Item 6
Item 7	Sub Item 7
Item 8	Sub Item 8
Item 9	Sub Item 9
Item 10	Sub Item 10

Admin Dashboard

Delete Database

Delete Requesters, Stock, Commands

DELETE

Reinitialize database

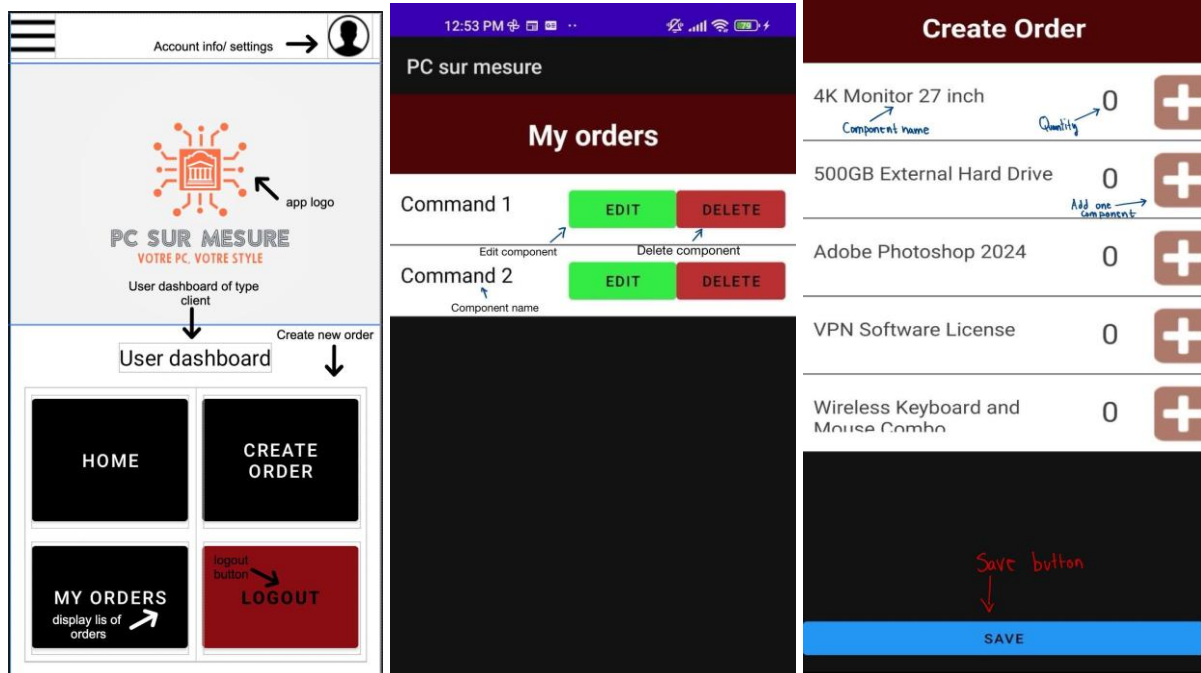
Override all Requesters and Stock

Reinitialize Stock

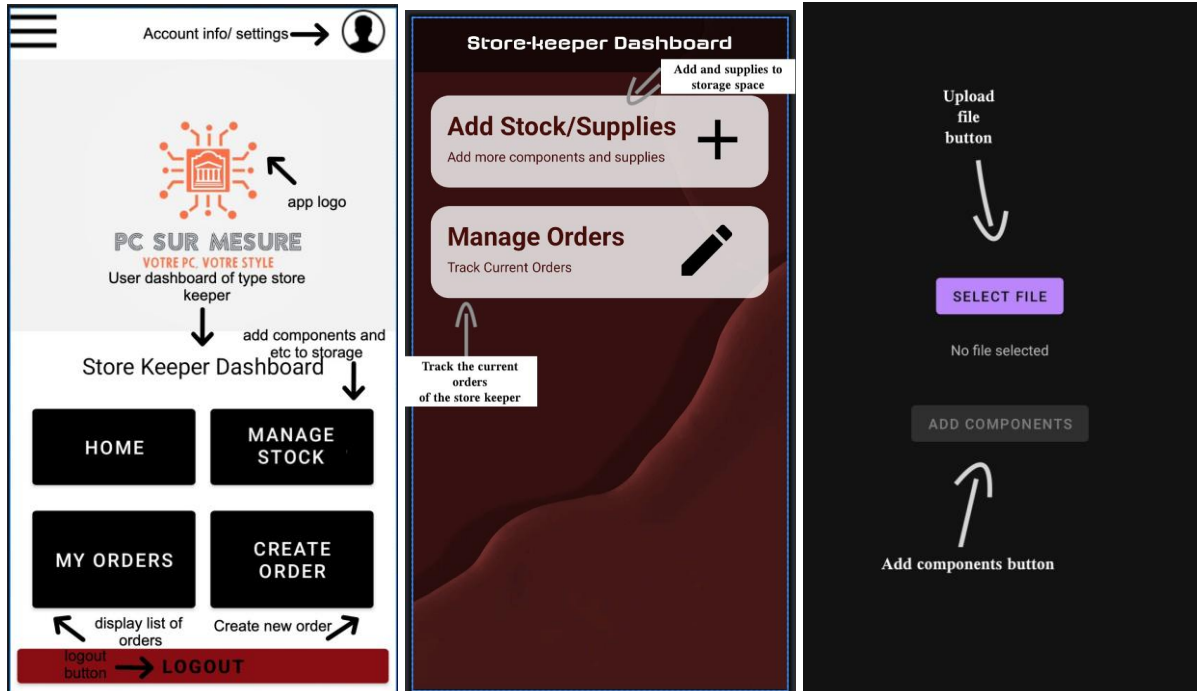
Override Stock

+

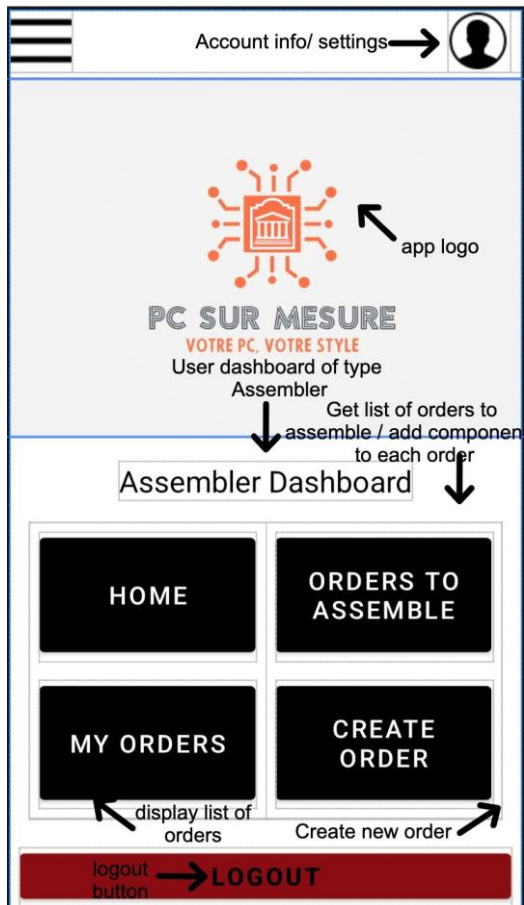
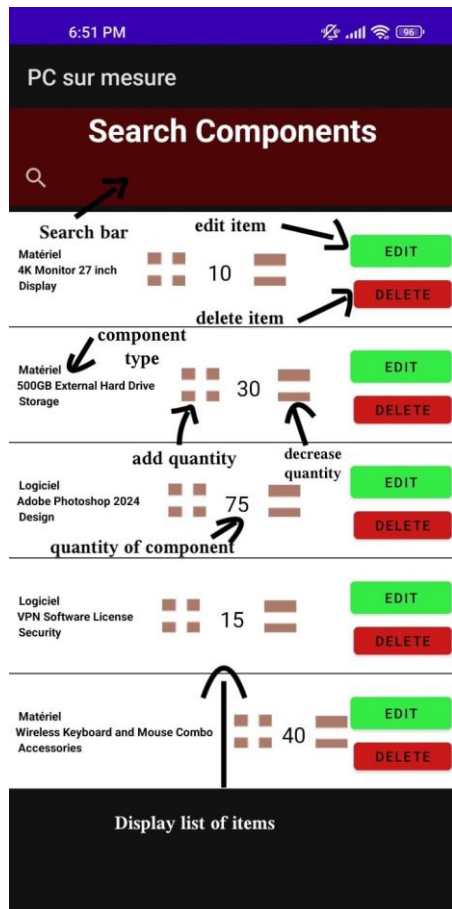
Client: Homepage



Storekeeper: Homepage



Assembler: Homepage



Description of Proposed or Handled Additional Requirements

Proposed and Handled Additional Requirements:

Advanced Search Functionality for Stock and Orders

Description: Added a search bar that allows users to filter stock components by type or sub-type.

Order Filtering: Users can also filter orders by their current status or the date they were created.

Implementation: Leveraging Firebase Firestore to execute complex queries. This feature significantly reduces the time required for users—especially the StoreKeeper—to manage a large inventory effectively.

Impact: Improves overall user productivity. Enhances the application's ease of use and accessibility.

User Action History

Description: Implemented a comprehensive logging system to record critical actions performed by users, such as creating new orders or modifying stock levels.

Implementation: These logs are stored directly in Firebase, providing a clear activity trail. This is particularly useful for the Administrator when conducting audits or troubleshooting system issues.

Impact: Ensures superior traceability across the platform. Facilitates the identification and resolution of potential errors or system abuse.

Discarded Additional Requirements

Integrated Chat System Between Roles

- **Description:** A proposed direct messaging system to facilitate communication between the various project users.
- **Reason for Rejection:**
 - High technical complexity that exceeded the initial project scope.
 - Availability of external communication tools (email, Slack, etc.) already utilized by the users.
- **Impact:** This functionality was deemed non-essential for the core ordering and assembly cycle.

Advanced PC Customization Option

- **Description:** A proposed interface allowing **Requesters** to configure detailed specifications, such as specific component performance metrics.
- **Reason for Rejection:**
 - Risk of excessive complexity for both inventory and order management.
 - Required frequent application updates to accurately reflect available stock.
- **Impact:** The current approach—selecting from predefined lists—remains simple, robust, and effective.

Overall Summary and Impact

These additional requirements, while limited in number, were carefully selected to maximize added value while respecting time and resource constraints. The implemented features (**Advanced Search** and **Action History**) significantly enhance the application's efficiency and interactivity, while the discarded features allowed the team to focus efforts on essential core elements.

Experience Feedback

Team Organization

From the project's outset, we prioritized structuring the team and establishing effective methods to meet deadlines while maintaining high product quality.

- **Role Distribution:** Roles were allocated based on each member's specific skills and interests:
 - **Requester Lead:** Management of the ordering system.
 - **StoreKeeper Lead:** Inventory management.
 - **Assembler Lead:** Oversight of assembly stages and delivered orders.
 - **Administrator Lead:** User management and database resets.
 - **Technical Coordinator:** Supervision of integrations and code reviews.
 - *Note: Every member also contributed to cross-functional tasks, including UML design, testing, and documentation.*
- **Collaboration Tools:**
 - **GitHub:** Utilized for version control, code reviews, and task organization through GitHub Projects.
 - **Discord:** Weekly meetings to discuss progress and resolve roadblocks.
- **Deliverable Planning:** The project was divided into four main stages (aligned with the deliverables), with intermediate milestones defined for each. We utilized "mini-deadlines" set before official dates to allow for a thorough revision period.

Challenges and Solutions

Issue: GitHub Conflict Management

- **Description:** Frequent conflicts occurred during merges, primarily when multiple members worked on the same files simultaneously.
- **Solution:** We adopted a strict naming convention for feature-based branches and implemented systematic code reviews before any merge. Additionally, we scheduled pair-integration sessions to minimize conflicts.

Issue: Firebase Integration Hurdles

- **Description:** During the implementation of Firebase for data management, several errors related to authentication and collection synchronization surfaced.
- **Solution:** We addressed this through extensive research of Firebase documentation and community forums, alongside targeted unit tests to validate data flows. One member was appointed as the Firebase lead to specialize in its advanced functionalities.

Issue: Role Coordination Difficulties

- **Description:** Certain roles were highly interdependent; for instance, the Assembler relied heavily on data provided by the StoreKeeper, which occasionally led to workflows being blocked.
- **Solution:** We organized pair programming sessions to quickly resolve these dependencies and establish a shared understanding of technical expectations.

Issue: Time Management

- **Description:** The project required a significant workload, and coordinating schedules was difficult due to the academic and personal obligations of various team members.
- **Solution:** We adopted a shared calendar and identified specific periods where extra resources were needed to compensate for member unavailability.

Lessons Learned for Future Projects

What Went Well:

- **Effective Collaboration:** Utilizing collaborative tools like GitHub maintained fluid communication and ensured clear task traceability.
- **Solid Initial Planning:** Defining roles and milestones at the start reduced unexpected issues and provided a clear project overview.
- **Proactive Problem Solving:** Weekly meetings helped us anticipate and quickly resolve technical or organizational obstacles.
- **Innovation and Creativity:** Adding extra features, such as advanced search for stock and orders, enriched the application and demonstrated our ability to exceed initial expectations.

Areas for Improvement:

- **Dependency Management:** We need to better anticipate interdependencies between roles earlier to avoid bottlenecks.
- **Test Enhancements:** Implementing more unit tests earlier in the process would have reduced regressions toward the end of the project.
- **Uniform Engagement:** Contribution levels were inconsistent among some members, leading to an uneven workload. Better individual task planning is needed to distribute efforts more equitably.

Team Brainstorming for Future Projects

- **Formalize Technical Documentation:** Establishing a clear guide for coding conventions and tool usage from day one would accelerate team integration.
- **Centralized Task Management:** Using tools like Jira would allow for even more precise management of responsibilities and deadlines.
- **Early Testing Phase:** Introduce unit and integration tests starting with the very first deliverable to detect anomalies immediately.
- **In-Person Brainstorming:** Organizing face-to-face sessions could strengthen team spirit and further reduce communication misunderstandings.

	Livable 1	Livable 2	Livable 3	Livable 4
Sofia El Ouazzani [300367668]	ui accountsettings ui signup ui login ui homepages rapport	Rapport ui storekeeper	Readme File Front Ui mesCommand es	Rapport Rapport finale
Youssef Cherkaoui [300360743]	ui homepages Drawer menu functionality Front end editUser method	Backend Frontend	Backend Frontend	Read me Frontend
Yasser El Mouatadir	-Set up firebase -Authentification -Add/removeUser r methods	-Readme	-Tests	-Backend -Tests
Amine Baba [300346890]	-UML -ReadMe -UI Adduser -UI remove user List	-3 UIs: ItemComponents Edit_remove_ad d_components	-3UIs: Creation_order Edit_order My_orders (et leurs listes)	2UIs: list orders waitlist list components of command
Ismail El Khayati	ui homepages ui remove user list ui sign up getAllusers	Frontend	Tests	Tests